

Anomaly fixing best practices

To fix the anomalies reported by TFDV in your dataset, you can take the following **systematic** approach:

1. Fix Data Type Mismatches

Error: "Expected data of type: INT but got FLOAT"

Fix:

- Convert these columns from float to int **after handling missing values:**

```
import pandas as pd
```

```
df = pd.read_csv("your_data.csv")
```

```
int_columns = ['age', 'fmlwgt', 'education_num', 'hours_per_week', 'capital_gain',  
'capital_loss']
```

```
for col in int_columns:
```

```
    df[col] = pd.to_numeric(df[col], errors='coerce') # convert to float first
```

```
    df[col] = df[col].fillna(0).astype(int)           # fill NA and cast to int
```

```
df.to_csv("your_data_fixed.csv", index=False)
```

2. Fix Unexpected String Values (e.g., "unknown", "??", "!")

Error: "Examples contain values missing from the schema: unknown (~2%)"

Fix:

1. **Standardize values** or **replace them with a placeholder**, or impute most frequent.

```
string_columns = [  
    'marital_status', 'sex', 'race', 'native_country', 'occupation',  
    'education', 'income', 'relationship', 'workclass'  
]
```

```
for col in string_columns:
```

```
    df[col] = df[col].replace(['unknown', '??', '###', '!', 'bad_data', 'missing', '123'], pd.NA)
```

```
    df[col] = df[col].fillna(df[col].mode()[0]) # or fill with a new category like 'other'
```

✓ If you want to keep 'unknown' as a valid category, you need to **update the schema** using TFDV to include that value.

3. Fix Distribution Drift

Error: "High Jensen-Shannon divergence between current and previous"

✓ **Fix:**

- This indicates significant drift compared to baseline.
- You can:
 - Accept the drift (update your baseline stats)
 - Investigate if drift is expected (e.g., real-world change)
 - Retrain models with updated data

```
# Recompute schema and stats
stats = tfdv.generate_statistics_from_dataframe(df)
schema = tfdv.infer_schema(stats)
tfdv.display_schema(schema)
```

```
# Or just update baseline stats
tfdv.write_stats_text(stats, "new_baseline_stats.txt")
```

Summary of Actions

Issue	Fix
FLOAT instead of INT	astype(int) after filling NaNs
Unexpected String Values	replace → fillna(mode) or update schema
Distribution Drift	Recompute baseline or investigate drift